

Python Tsunami

Local Installation

Local Installation Options



- JupyterLab
- **Visual Studio Code (VSC)**
- PyCharm
- Spyder

<https://code.visualstudio.com/>



Visual Studio Code



- Free + open source solution from Microsoft
- Many programming languages
- Github integration
- Easy to install



<https://code.visualstudio.com/>

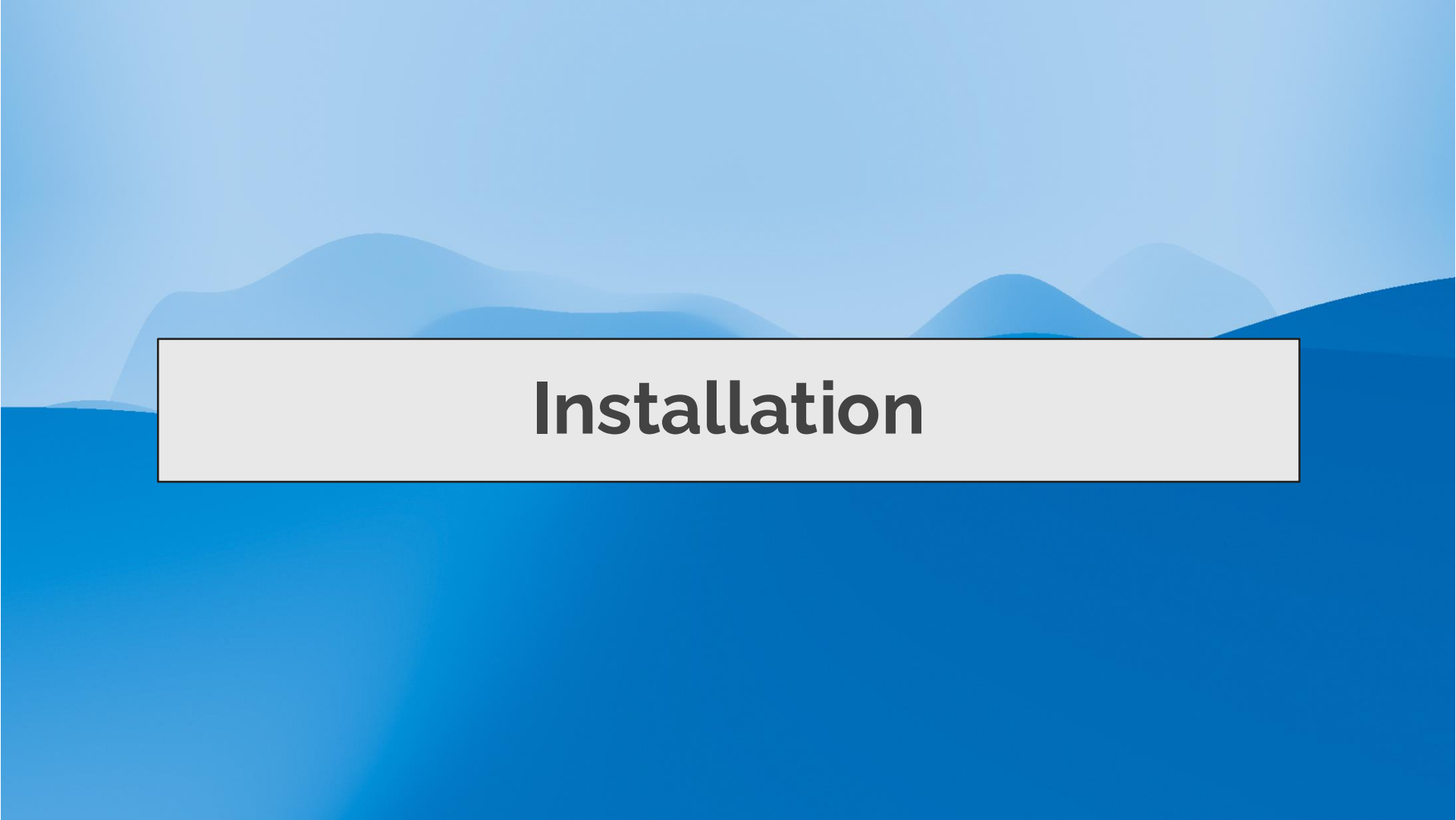
Disable telemetry if you would not like Microsoft collecting your data!

How to disable telemetry reporting

VS Code collects usage data and sends it to Microsoft to help improve our products and services. Read our [privacy statement](#) and [telemetry documentation](#) to learn more.

If you don't want to send usage data to Microsoft, you can set the `telemetry.telemetryLevel` user [setting](#) to `off`.

From **File > Preferences (Code > Preferences or Code > Settings on macOS) > Settings**, search for `telemetry`, and set the **Telemetry: Telemetry Level** setting to `off`. This will silence all telemetry events from VS Code going forward.

The background of the slide features a stylized landscape with rolling hills or mountains in various shades of blue. The foreground is a solid, vibrant blue, while the background hills are rendered in lighter, semi-transparent blue tones, creating a sense of depth. A white rectangular box with a thin black border is centered horizontally and positioned in the upper-middle part of the slide.

Installation

Installing Python and Libraries - OS X & Ubuntu



- Install Python via system package manager or official installer:
 - homebrew: <https://docs.brew.sh/Homebrew-and-Python>
 - apt-get: <https://help.ubuntu.com/community/AptGet/Howto>
- Install Visual Studio Code:
 - <https://code.visualstudio.com/docs/setup/mac>
 - Add the official Python & Jupyter extension in Visual studio code
- Set up a Python environment for reproducibility (venv or Conda)
- Install required libraries
 - Pip / Pip3: <https://www.activestate.com/resources/quick-reads/how-to-install-and-use-pip3/>
 - Conda: <https://docs.conda.io/projects/conda/en/stable/user-guide/concepts/installing-with-conda.html>



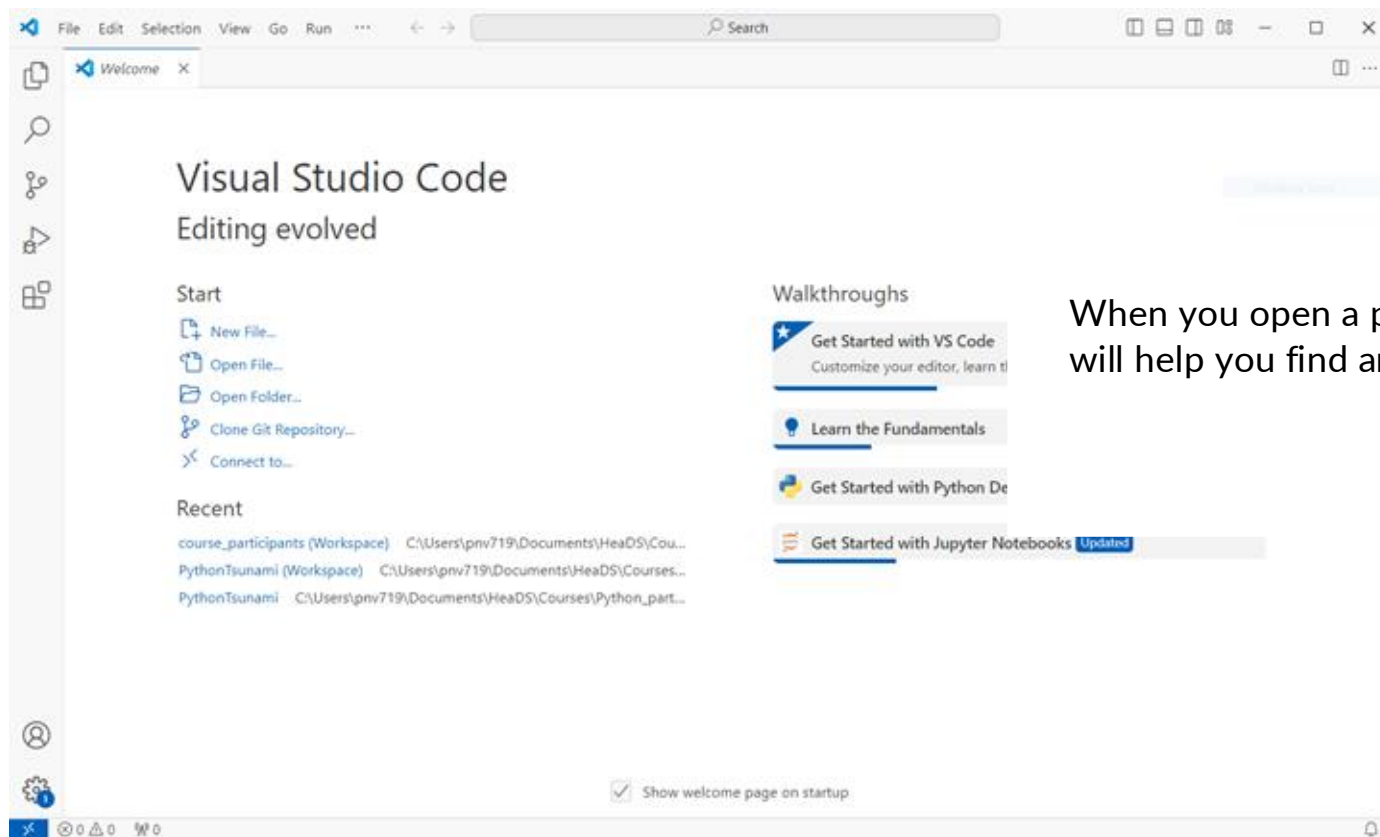
Installing Python and Libraries - Windows



- **Install Python for Windows**
 - Use the official Python installer: <https://www.python.org/downloads/windows/>
 - Ensure Python is added to system PATH (ask chatGPT)
- **Install Visual Studio Code:**
 - <https://code.visualstudio.com/docs/setup/windows>
 - Add the official Python & Jupyter extension to VS code
- **Set up a Python environment for reproducibility (venv or Conda)**
- **Install required libraries**
 - Pip: <https://packaging.python.org/en/latest/tutorials/installing-packages/>
 - Conda: <https://docs.conda.io/projects/conda/en/stable/user-guide/concepts/installing-with-conda.html>

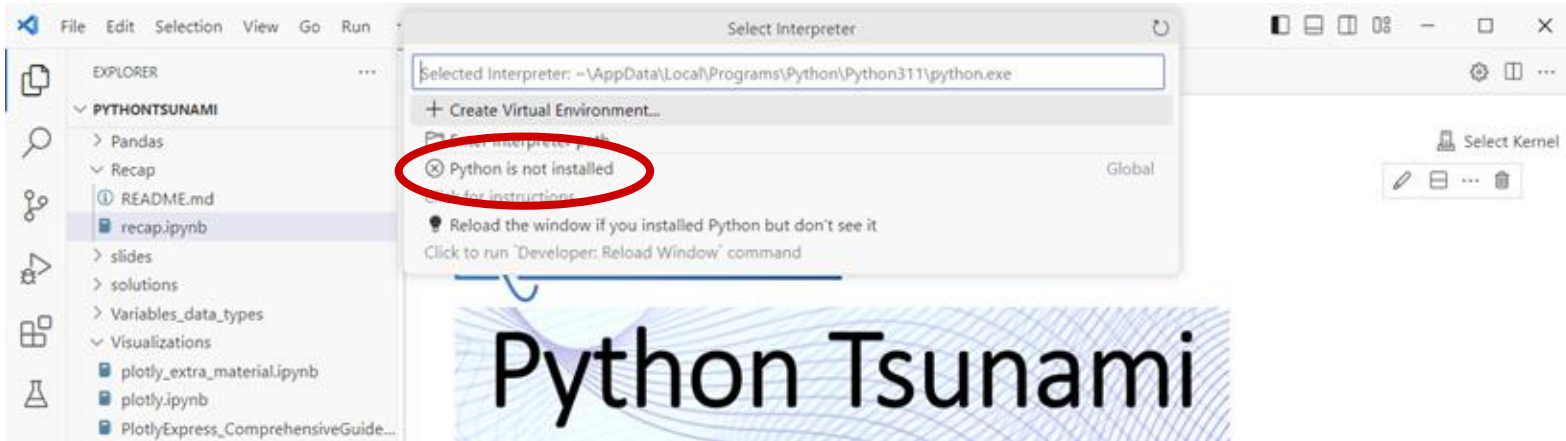


Start Up VS code



When you open a python script it will help you find an interpreter

Add Python Extension in VS code

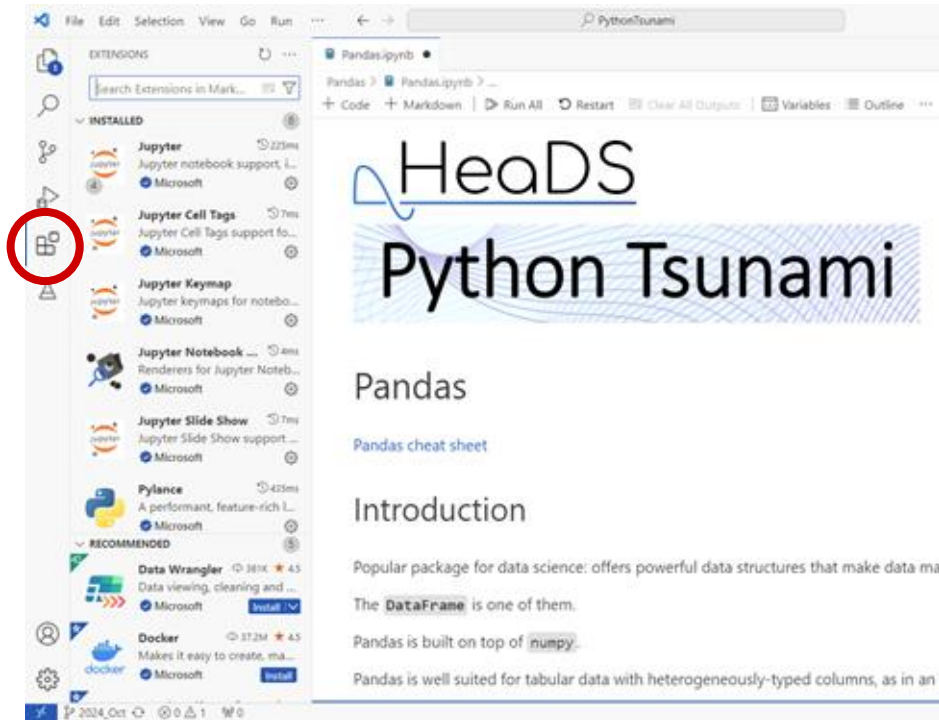


Windows: Microsoft Store or global install <https://www.python.org/downloads/windows/>

Mac: App Store or command line

Linux systems: Command line

Add Python Extension in VS code



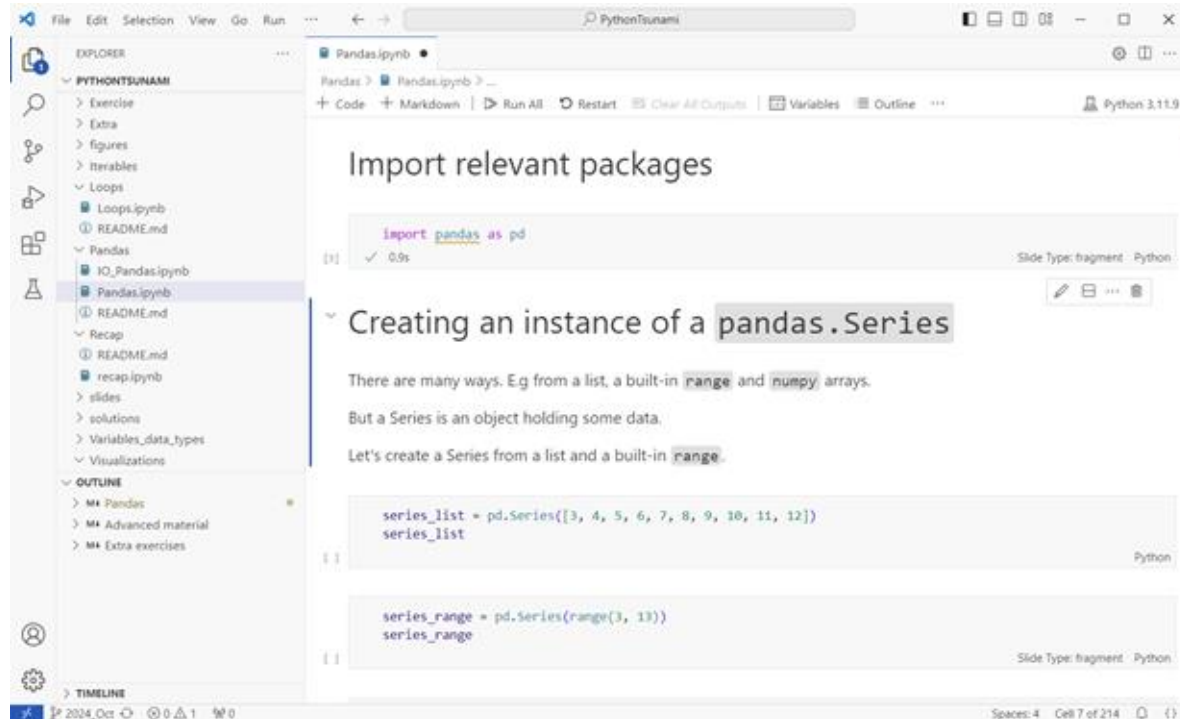
OR check 'extensions'

As you can see, I also got Jupyter ect. because I opened an ipython notebook.

Jupyter Notebook



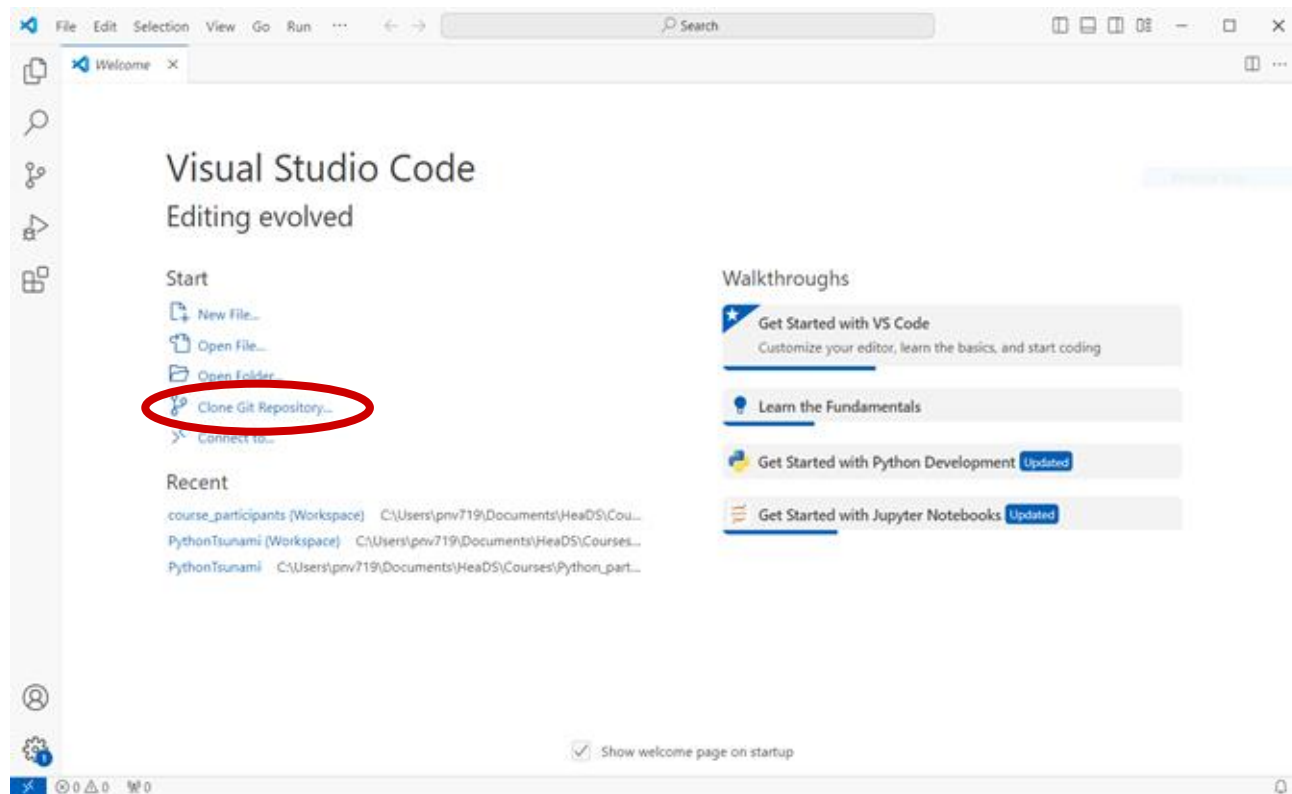
VS code can open ipython notebooks:



Operate as in colab

- Code cells and text cells
- Run cells with run button or shift+enter
- Output appears below cell

Github Integration



Packages



Packages are also called libraries.

They are bundles of code that contain a certain functionality.

You have already used packages:

```
import pandas as pd
```

```
import seaborn as sns
```

You'll need to have a local install of the package before you can load and use it.



Installing Python and Libraries



Option 1: Pip + any bash terminal

```
MINGW64/c:/Users/pnv719
pnv719@SUN1022949 MINGW64 ~
$ pip install pandas
Collecting pandas
  Downloading pandas-2.2.2-cp311-cp311-win_amd64.whl.metadata (19 kB)
Collecting numpy>=1.23.2 (from pandas)
  Downloading numpy-2.1.1-cp311-cp311-win_amd64.whl.metadata (59 kB)
----- 59.7/59.7 kB 1.5 MB/s eta 0:00:00
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\pnv719\appdata\local\packages\pythonsoftwarefoundation.python.3.11_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from pandas) (2.9.0.post0)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2024.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Downloading tzdata-2024.1-py2.py3-none-any.whl.metadata (1.4 kB)
Requirement already satisfied: six>=1.5 in c:\users\pnv719\appdata\local\packages\pythonsoftwarefoundation.python.3.11_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Downloading pandas-2.2.2-cp311-cp311-win_amd64.whl (11.6 MB)
----- 11.6/11.6 MB 13.6 MB/s eta 0:00:00
Downloading numpy-2.1.1-cp311-cp311-win_amd64.whl (12.9 MB)
----- 12.9/12.9 MB 16.4 MB/s eta 0:00:00
Downloading pytz-2024.2-py2.py3-none-any.whl (508 kB)
----- 508.0/508.0 kB 15.6 MB/s eta 0:00:00
```

Option 2: Pip in powershell

```
Command Prompt
Microsoft Windows [Version 10.0.19045.4894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pnv719>pip.exe install mechanize
Collecting mechanize
  Downloading mechanize-0.4.10-py2.py3-none-any.whl.metadata (4.3 kB)
Collecting html5lib>=0.999999999 (from mechanize)
  Downloading html5lib-1.1-py2.py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: six>=1.9 in c:\users\pnv719\appdata\local\packages\pythonsoftwarefoundation.python.3.11_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from html5lib>=0.999999999->mechanize) (1.16.0)
Collecting webencodings (from html5lib>=0.999999999->mechanize)
  Downloading webencodings-0.5.1-py2.py3-none-any.whl.metadata (2.1 kB)
Downloading mechanize-0.4.10-py2.py3-none-any.whl (110 kB)
----- 110.4/110.4 kB 2.1 MB/s eta 0:00:00
Downloading html5lib-1.1-py2.py3-none-any.whl (112 kB)
----- 112.7/112.7 kB 6.4 MB/s eta 0:00:00
Downloading webencodings-0.5.1-py2.py3-none-any.whl (11 kB)
Installing collected packages: webencodings, html5lib, mechanize
Successfully installed html5lib-1.1 mechanize-0.4.10 webencodings-0.5.1

[notice] A new release of pip is available: 24.0 -> 24.2
[notice] To update, run: C:\Users\pnv719\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip

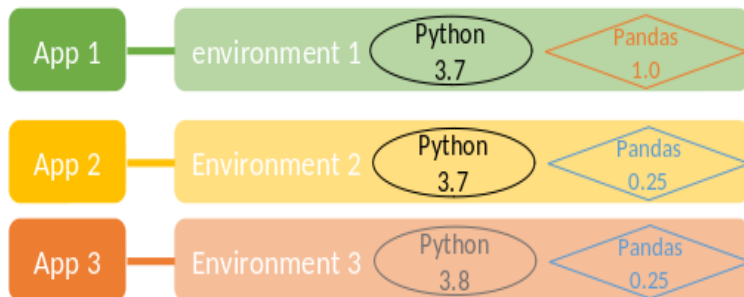
C:\Users\pnv719>
```

Some systems have pip2 or pip3 instead of pip.

Conda Environment



- **Conda** is an open source package & environment management system.
- Softwares exist as conda packages, you can use conda to install them and their dependencies.
- **An environment** is a specific combination of packages in a specific version.



- PROS:** Comprehensive, many options
Environment management
Well documented
- CONS:** Can be overwhelming
Some channels are **not** free

Dataset Exercise

